# Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

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For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

## 0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

The programming concepts from this class that will help me complete the assignment are lists, different loops, dataclasses, as well as Drafter implementation like State, and buttons, pages, and routes. I want to improve my Drafter knowledge and programming through this project.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

I think the easiest part of the projects will be designing the dataclasses for the different things I want to track. This is just assigning dataclasses with variables and data types. The hardest will be connecting everything together in Drafter to actually make the buttons function, and carry data over from different pages. I do not have as much experience in Drafter as I do with writing functions and dataclasses.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

I think the Drafter pages and actual math to calculate different things in the site might take the longest and require the most thought and effort. I think figuring out the basic dataclasses and paths between pages is the best first step, and then I can flush out each page with the required information.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

My phone and other assignments will definitely be distracting during my work. I can minimize them by turning my phone off or keeping it away from me, as well as completing my other class work in a timely manner to leave time to work on the project. While I'm working, I will need to have my computer, Thonny, and previous work on Canvas to help me out.

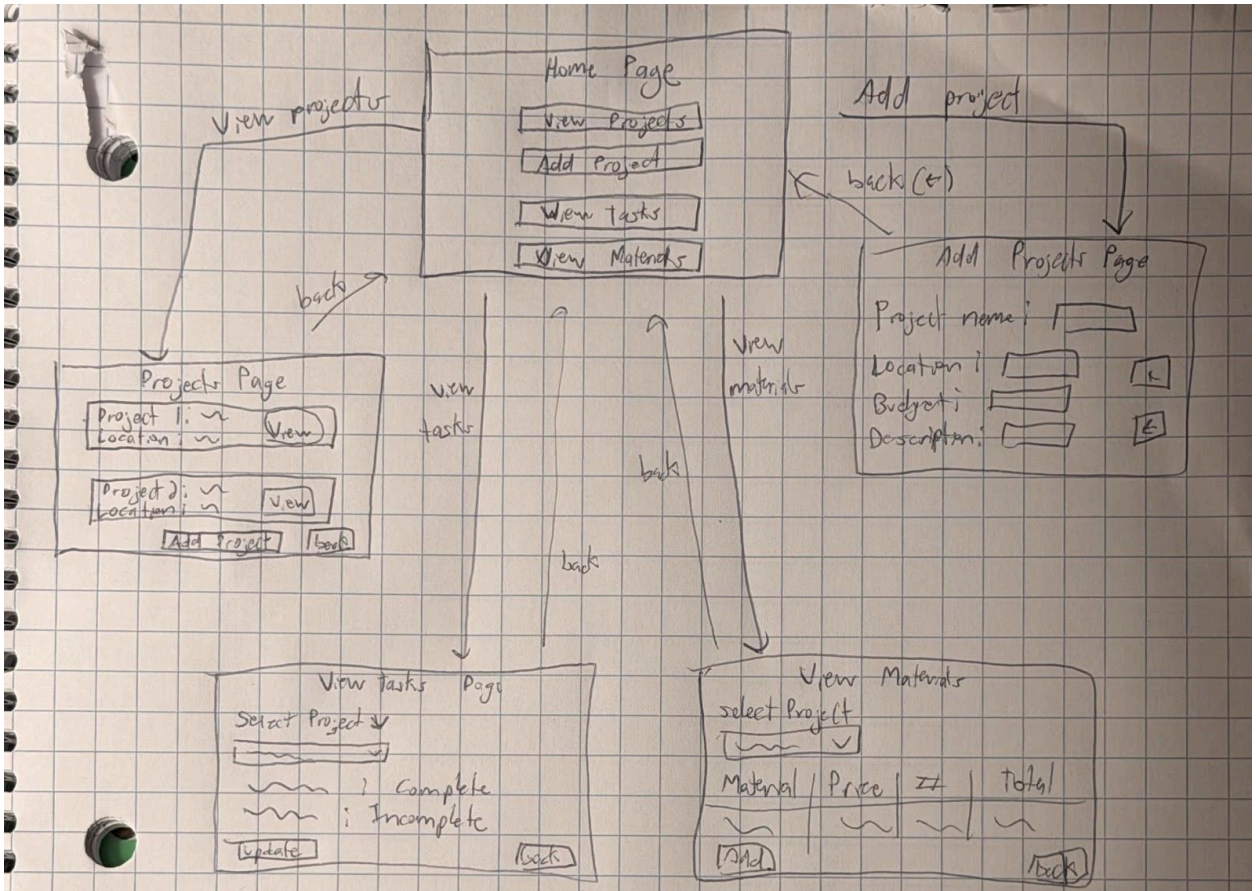
# 1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.




State dataclass

projects : list [Projects]
tasks : list [tasks]
materials : list [materials]
selected_project: str
page_message : str

# 2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

| **Requirement** | **Implementation** |
|---|---|
| **Purpose/Functionality**<br><br>A 1-2 sentence description of the purpose or functionality of your site | **My website will help civil engineers and construction managers to organize their ongoing projects, as well as track materials and costs.** |
| **State Dataclass Attributes**<br><br>Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass.<br><br>Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site. | projects: List of Project instances<br>tasks: list of Task instances<br>materials: list of Material instances<br>Selected_project: string (display selected project)<br>page_message: string (message shown to the user)<br><br>Projects will store the construction projects, and will add new ones when the user makes them<br>Tasks also store the tasks, and will be modified to add whatever the user inputs<br>Materials will be modified by a user input adding or removing them<br>Selected_project will change as user selects different projects<br>Page_message will display different messages depending on what actions the user takes |
| **Second Dataclass**<br><br>Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified) | Project:<br>project_name: str (stores the name of a specific project, and is made by the user)<br>location: str (entered by user and stored as a location for a project)<br>budget: float (entered by the user, stores a budget for |

| | |
|---|---|
| | the project)<br>description: str (describes what the project is) |
| **Routes** (minimum 7)<br><br>The names of your route functions | home(), projects(), add_project()<br>tasks(), add_task(), materials(),<br>add_material(), project_details(), update_task_status() |
| **Pages** (minimum 5)<br><br>A brief description of the pages in your site | Home page<br>Projects page<br>Add Project page<br>Tasks page<br>Materials page<br>Project Details page |
| **User Input** (minimum 3)<br><br>Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes<br><br>For more information on these input types, see Drafter Documentation. | TextBox for project name, project location, project budget, task name, material name, material amount, material price<br><br>SelectBox for tasks |
| **If Statements** (minimum 3)<br>Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site | 1. One if statement will check if project names are empty before creating a project with inputted name<br>2. Another one will check if a task in complete when determining how much progress has been made<br>3. A third one will check the input of a material price before using it in calculations to make sure it is a valid input |
| **Loop** (minimum 1) | I will use a for loop to loop through a list of Project instances, which will be stored in projects of the State |

| | |
|---|---|
| This loop must iterate over the list/dict of dataclass instances in your State attribute.<br>Description of this loop and why it is meaningful to the site | dataclass. It will allow me to display the information about each project in the list, without having to input each one one at a time.<br>It is meaningful because it makes it easiest to display the information about all ongoing projects on the website.<br>There will be more loops for other dataclass attributes as well |

# LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in *this* assignment, simply say so below.

***do you remember the civil engineering website I was planning?***

*Yes! We were planning a Construction Project Manager website for your Computer Science project, using Python and Drafter.*

*We actually went through a couple of ideas before settling on the final concept.*

*Your website: Construction Project Manager*

*The goal was to create a website that helps manage construction projects, track tasks, and monitor material costs.*

***what are clear and accurate names for routes I will have in my program?***

*Based on your Construction Project Manager idea, we can use your existing concept to determine effective route names.*

*Routes*

`home()`, `projects()`, `add_project()`, `tasks()`, `add_task()`, `materials()`, `add_material()`, `project_details()`, `update_task_status()`.

# Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.